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AWS CLOUD INFRASTRUCTURE IMPLEMENTATION REPORT

Secure E-Commerce Application (Myntra) Architecture

1. Amazon EC2 – Compute Layer

The EC2 console shows three instances in the Asia Pacific (Mumbai) region. Two instances are running and one is stopped. The visible instance types are t3.micro and t3.small, indicating a lightweight deployment suitable for development, testing, student projects, or small workloads.

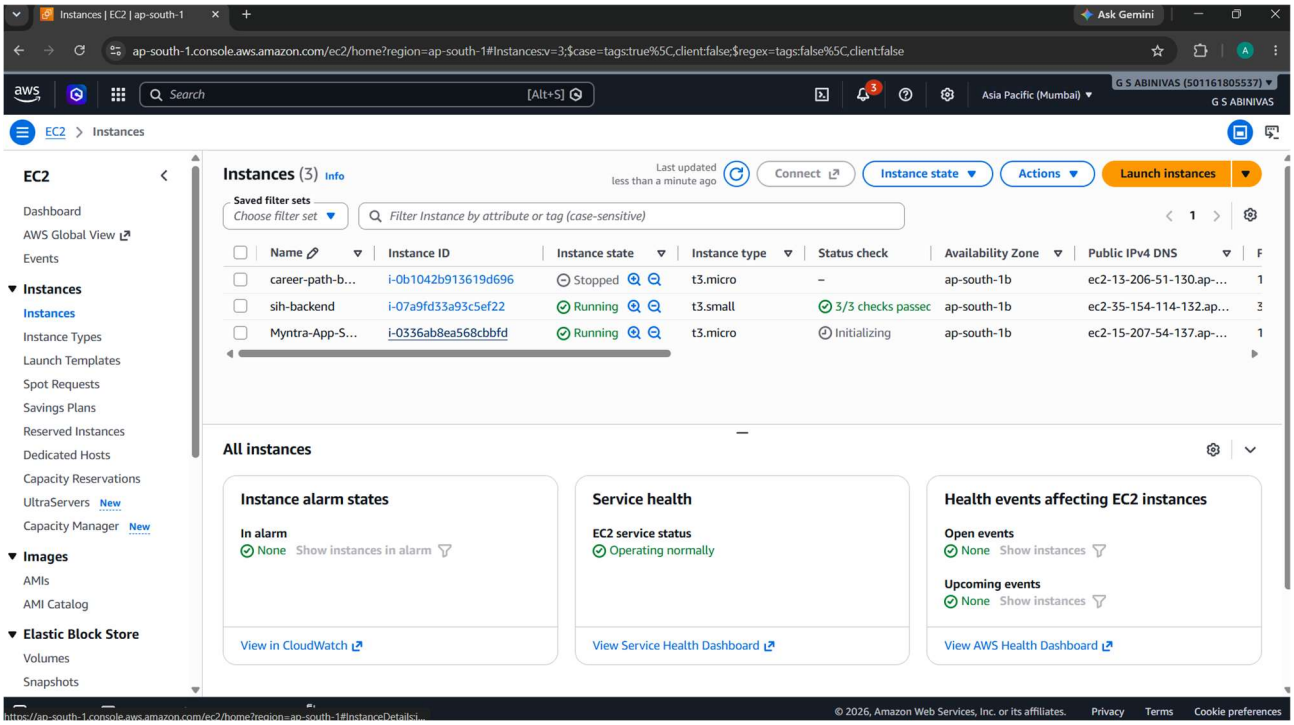


Figure 1. EC2 instances observed in ap-south-1.

2. Amazon S3 – Object Storage

The S3 console shows a general-purpose bucket named myntra-product-images in ap-south-1. The naming strongly indicates that the bucket is intended to store product image assets for the application. The screenshot also exposes AWS's external-access summary area, which is relevant for checking whether bucket access could unintentionally become public or cross-account.

Resource	Observed configuration	Recommended control
Bucket	myntra-product-images	Use least-privilege bucket policy/IAM access
Region	ap-south-1	Keep compute and storage co-located when practical
Data type	Product images	Define lifecycle and retention policy as needed

Access	External access summary visible	Verify Block Public Access and bucket policy

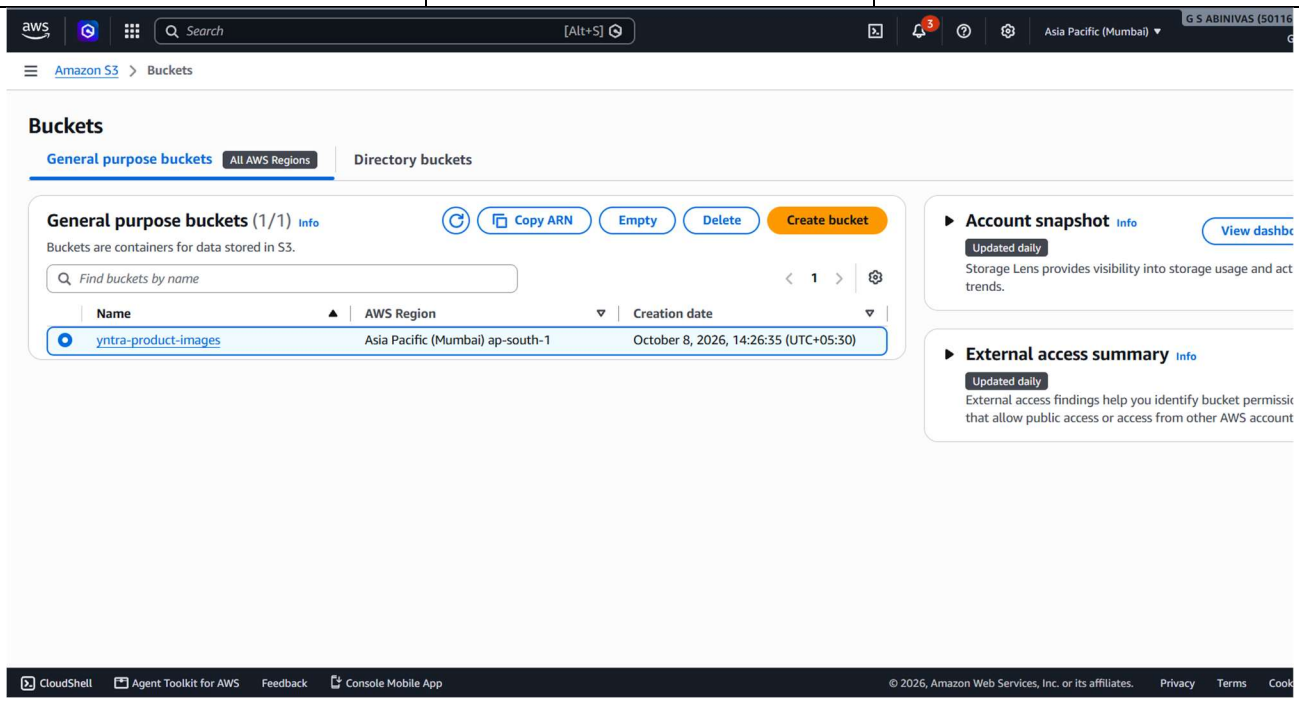


Figure 2. S3 product-image bucket.

2. AWS IAM – Identity and Access Management

IAM evidence shows a combination of AWS service-linked roles, custom/service roles, IAM users, and IAM groups. This indicates that the environment is using multiple identity mechanisms rather than relying only on the root account.

2.1 IAM Roles

The Roles page shows nine roles in the captured account. Visible examples include AWS service-linked roles and custom roles such as career-path-backend-bedrock-role, Myntra-EC2-S3-Access-Role, and a monitoring-related role. These roles are important because EC2 applications can access AWS services without embedding long-lived access keys directly in application code.

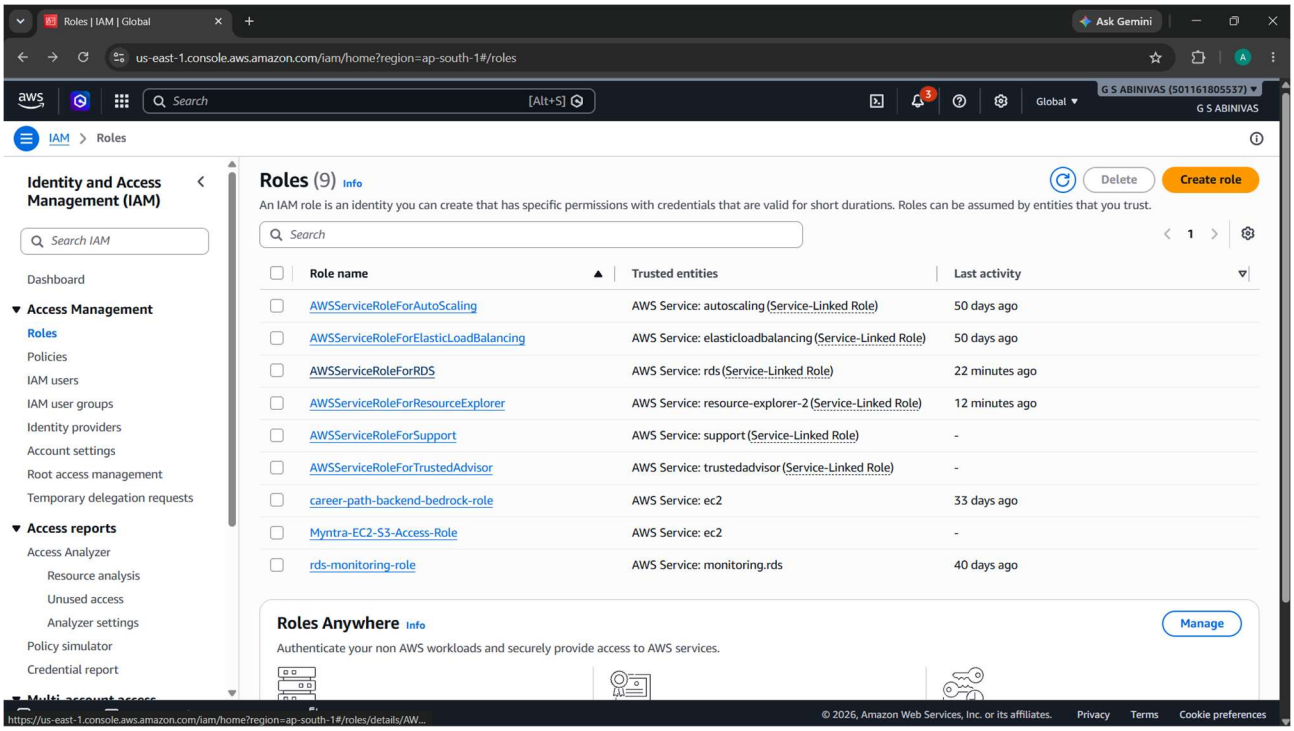


Figure 3. IAM roles observed.

2.2 IAM Users

Five IAM users are visible. The screenshot includes users associated with application/API access, development/testing, and other account activities. Long-lived IAM users should be minimized for workloads where IAM roles or federated identities can be used instead.

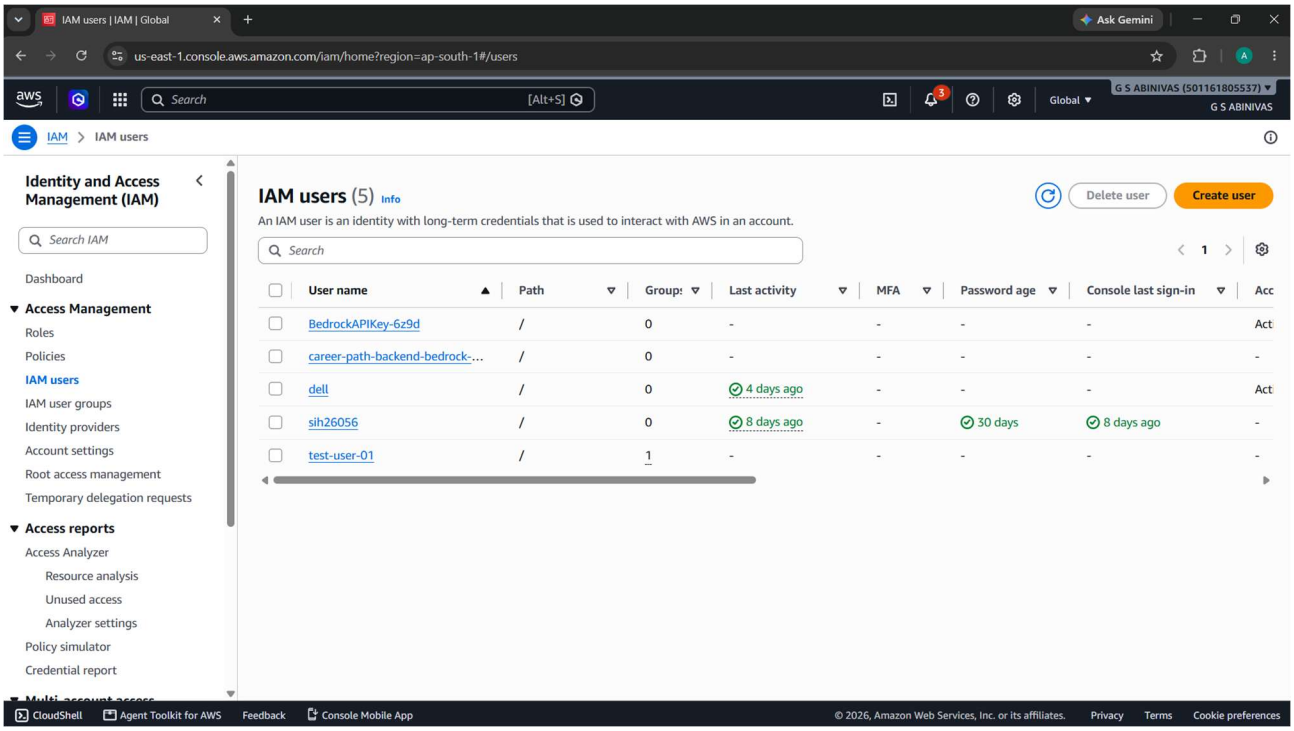


Figure 4. IAM users observed.

2.3 IAM User Groups

Two IAM groups are visible: Myntra-Developers and Myntra-Testers. Groups provide a scalable mechanism for assigning common permissions to multiple users and are preferable to manually duplicating policies user by user.

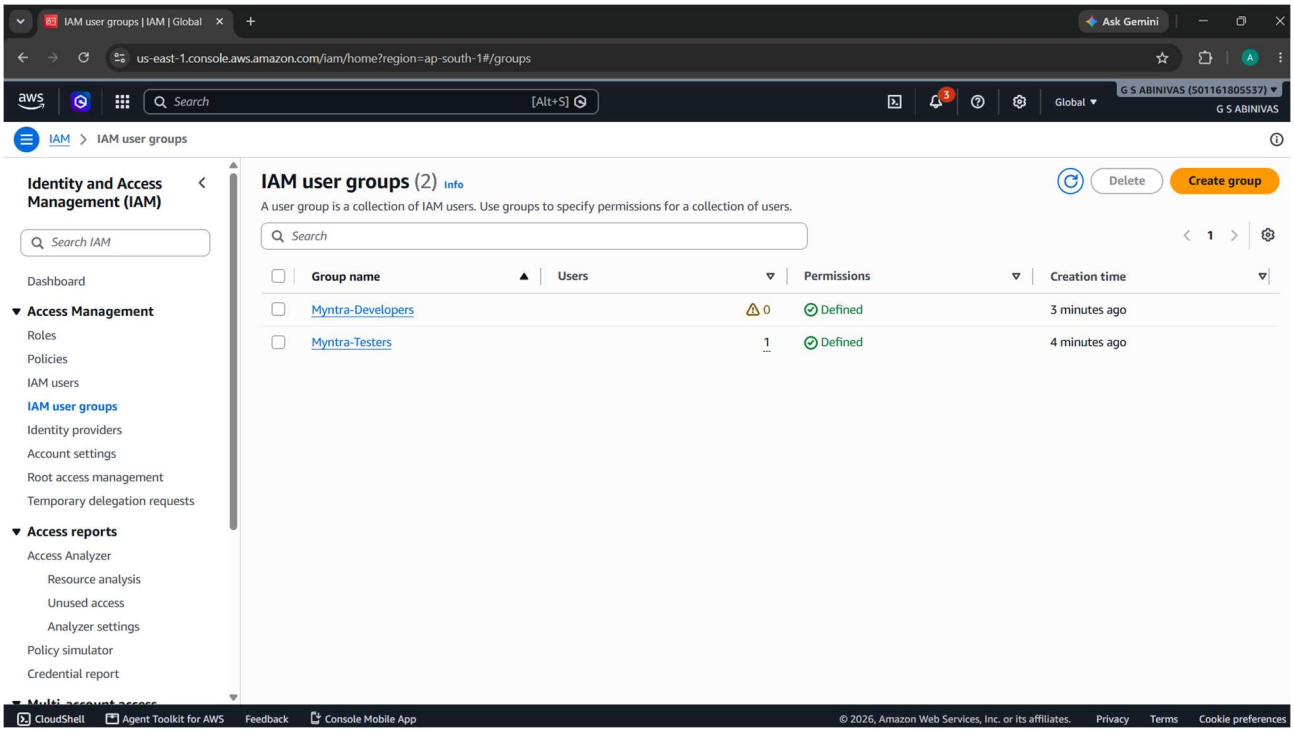


Figure 5. IAM user groups observed.

3. Network Security – Security Group

The security group myntra-web-sg is shown as being created successfully. The visible inbound rules allow HTTP on port 80, HTTPS on port 443, and SSH on port 22. The screenshot also shows one outbound rule entry. The exact source CIDRs are not fully legible in the supplied capture, so they should be verified directly in AWS before treating the rules as production-ready.

Port	Protocol	Service	Purpose	Security note
80	TCP	HTTP	Web traffic	Prefer redirecting HTTP to HTTPS
443	TCP	HTTPS	Secure web traffic	Primary public web endpoint
22	TCP	SSH	Administrative access	Restrict source IPs or use SSM where possible

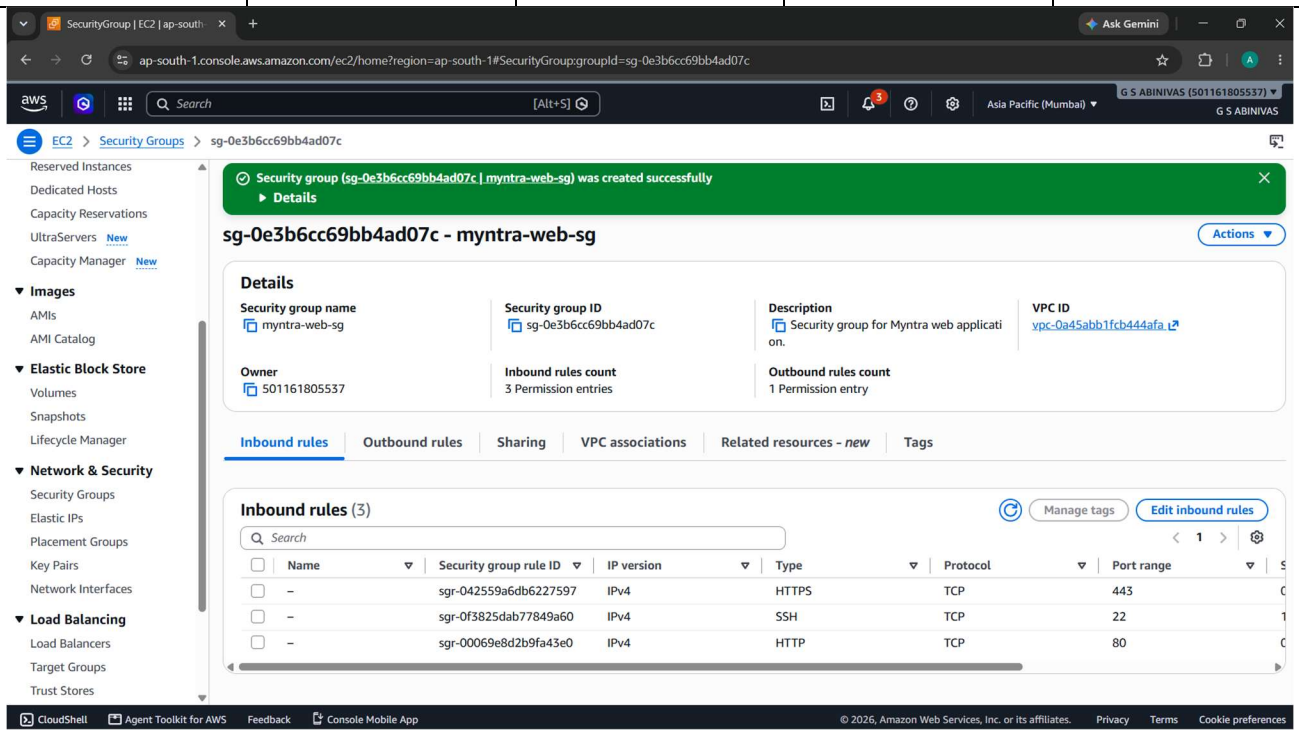


Figure 6. myntra-web-sg security group and inbound rules.

4. End-to-End Application Flow

1	**User / client request**	A client reaches the web application through the network entry point controlled by the EC2 security group.
2	**EC2 application layer**	The request is handled by an application or backend workload hosted on an EC2 instance.
3	**IAM authorization**	When the workload needs AWS resources, an attached IAM role can provide temporary permissions without hard-coding access keys.
4	**S3 object access**	Product images are stored or retrieved from the myntra-product-images S3 bucket.
5	**Response**	The EC2-hosted application returns the requested application response or asset to the client.

This flow is the most defensible interpretation of the supplied evidence. It describes the relationship between the visible resources without inventing hidden application components.

5. Security and Operational Assessment

Area	**Current evidence**	**Assessment / next action**
Identity	Roles + groups + users present	Good foundation; enforce least privilege and MFA
EC2 access	SSH : 22 visible	Restrict source ranges; consider AWS Systems Manager
Web access	HTTP : 80 and HTTPS : 443	Use HTTPS as primary path and redirect HTTP
S3	Dedicated product-image bucket	Verify Block Public Access, encryption and bucket policy
Compute	3 instances, 2 running	Add monitoring, patching and backup procedures
High availability	Single AZ visible	For production, evaluate multi-AZ architecture
Scaling	No scaling configuration visible	Consider Auto Scaling if workload requires it
Observability	CloudWatch links visible in EC2 console	Define CPU, memory/application, error and availability metrics

13. Conclusion

The supplied AWS evidence demonstrates a functional cloud foundation centered on EC2 compute, S3 object storage, IAM access management, and EC2 security groups. The environment is structured around application workloads hosted in the Mumbai region, with a dedicated S3 bucket for product images and explicit identity/network controls.

For a production-grade deployment, the highest-priority improvements are identity hardening, tighter SSH access, S3 access validation, HTTPS enforcement, monitoring, and—if availability requirements justify it—multi-AZ and load-balanced architecture.

Appendix A – Evidence Index

Figure	Screenshot	Evidence
Figure 1	Screenshot 2026-10-08 143734.png	Amazon EC2 – Instances
Figure 2	Screenshot 2026-10-08 142713.png	Amazon S3 – Product Image Storage
Figure 3	Screenshot 2026-10-08 143235.png	AWS IAM – Roles
Figure 4	Screenshot 2026-10-08 143246.png	AWS IAM – Users
Figure 5	Screenshot 2026-10-08 143255.png	AWS IAM – User Groups
Figure 6	Screenshot 2026-10-08 143446.png	Amazon EC2 – Security Group